NOTE: The instructor reserves the right to update small details and undetermined features of this syllabus during the semester. Changes will be announced in class and posted on Brightspace.

MATH 547: Mathematical Foundations of Statistical Learning Theory

Units: 3
Term–Day–Time: Fall 2026; Monday, Wednesday, and Friday; 1:00–1:50 PM
Location and Modality: KAP 140; in person

Instructor Information

Instructor: Manuel Fernandez V
Office: Room 464B
Office Hours: Friday, 2:00–3:00 PM, Room 464B
Contact: manuelf7@usc.edu. Use the subject line "MATH 547 FALL 2026 [student: insert name]." Messages received by 9:00 AM on a business day will generally receive a response by 5:00 PM that day; later and weekend messages roll over to the next business day.

Teaching Assistant Information

There is no teaching assistant assigned to this course.

Course Description

This course introduces the mathematical foundations of *Statistical Learning Theory*, together with selected topics in high-dimensional statistics and high-dimensional probability. The course emphasizes rigorous mathematical formulation, proof techniques, and the relationship between learning-theoretic guarantees and statistical methodology.

Prerequisites

Working knowledge at the graduate or advanced undergraduate level of probability theory (MATH 407/505a or 507a recommended), real analysis (MATH 425a/b or 525b recommended), and linear algebra.

Course Notes and Technology

Lecture notes and announcements will be posted on Brightspace. The course has no regular Zoom component unless the instructor must teach remotely. No specialized hardware or software is required beyond a PDF reader, a LaTeX installation or editor, and access to Brightspace.

Readings and Supplementary Materials

Notes and Excerpts

The class will be taught using notes provided by the instructor. Relevant excerpts from the sources listed below will be posted each week on Brightspace. Students are responsible for the posted notes and excerpts; no single textbook is required.

Optional references

- **[UML]** Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. The authors provide a free electronic version.
- **[K]** Vladimir Koltchinskii, *Oracle Inequalities in Empirical Risk Minimization and Sparse Recovery Problems*.
- **[HDP]** Roman Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, second-edition draft.
- **[Tao]** Terence Tao, *Topics in Random Matrix Theory*.
- **[LV]** Yin Tat Lee and Santosh Vempala, *Techniques in Optimization and Sampling*, August 7, 2026 draft.
- **[GN]** Evarist Giné and Richard Nickl, *Mathematical Foundations of Infinite-Dimensional Statistical Models*.

Assessments and Assignments

Weekly Practice Problems

Problems based on the material taught in class will be posted each week. Students should solve them as preparation for the midterm examinations and, if they choose to take it, the final examination. Each set may be submitted during the two-week window beginning when the problems are posted.

Practice-problem submissions do not carry a percentage of the course grade and will not be graded. The instructor may, however, consider a student's submitted work as extra credit if that student seeks an improved grade at the end of the semester. Poorly written or largely incorrect solutions will not be considered for that purpose. Collaboration is allowed, but each student must write their own solutions and acknowledge all collaborators. **Submitted solutions must be written in LaTeX, and both the PDF and source .tex file must be submitted.**

In-class midterm examinations

There will be two in-class midterm examinations. Each midterm is worth 20% of the final course grade.

Final exam or finalproject

The final summative assessment is worth 60% of the final course grade. Students may complete either a comprehensive final examination or a finalproject. The instructor will announce the project requirements and milestones during the semester. The university-assigned final period is **Wednesday, December 16, 2026, 11:00 AM–1:00 PM**. If a project is used, the final submission or presentation will occur during that period unless the instructor announces a policy-compliant alternative.

Grading Breakdown and Final Scale

Table 1. Grading breakdown

Assessment	Format	Percentage
Midterm examination 1	In class	20%
Midterm examination 2	In class	20%
Weekly practice problems	Ungraded; possible extra-credit consideration	0%
Final exam or finalproject	Student choice	60%
Total		**100%**

Final grading scale

No fixed letter-grade scale is set in advance. The instructor will determine the final grading scale at the end of the semester based on the overall performance of the class. The purpose of the course is to learn, and it should not be hard to earn a good grade. It is certainly possible for every student in the class to receive an A, and it is unlikely—although not impossible—for any student to receive less than a B+.

Submission, Feedback, and Course Policies

Practice-problem submission

Practice problems will be submitted through Brightspace. The submission window for each weekly set will remain open for two weeks after that set is posted. Submissions received after the window closes will not normally be considered for extra credit unless an extension has been approved in advance.

Attendance

Attendance is not mandatory, but it is strongly encouraged. Students remain responsible for announcements, material, and in-class work missed during an absence. University-approved absences will be accommodated consistently with USC policy.

Use of artificial intelligence tools

Students may use generative artificial intelligence tools to help solve weekly practice problems. Any such submission must identify the AI model and version used. The student remains responsible for understanding, checking, and being able to explain the resulting work.
Students may not use AI while taking either midterm examination or the final examination. Students who decide to do a final project may not use AI to draft, rewrite, typeset, or otherwise write down any part of it. AI systems commonly produce LaTeX in recognizable ways, so prohibited use may be relatively apparent to the instructor. A student found to have used AI in any prohibited assessment will automatically receive an F in the course. A student who is concerned about complying with the finalproject restriction should feel free to take the final examination instead.

Course content and recordings

Recording a class without the express permission of the instructor and notice to the class is prohibited unless conducted pursuant to an Office of Student Accessibility Services accommodation. Distribution or commercial use of notes, recordings, examinations, solutions, or other course materials without the instructor's permission is also prohibited. See the USC Student Handbook.

Academic Integrity

The University of Southern California is a learning community committed to the pursuit and transmission of knowledge. Academic misconduct is contrary to that mission and includes dishonesty in submitted work; cheating; plagiarism; fabrication; unauthorized assistance; knowingly assisting another person in academic dishonesty; and any act intended to gain an unfair academic advantage. Students are expected to uphold the highest standards of academic integrity in all course work.

All submitted work must be original and prepared specifically for this course in this academic term. Work written by others or recycled from another course may not be submitted without written permission. Suspected misconduct will be reported to the Office of Academic Integrity. Violations may result in a grade penalty and university disciplinary action. Consult the Student Handbook, the Office of Academic Integrity, and the university's policy on Research and Scholarship Misconduct. Ask the instructor whenever the boundary between authorized and unauthorized assistance is unclear.

Learning Experience Evaluations

USC learning experience evaluations provide an opportunity to give feedback on the course and instruction. Students are encouraged to participate in the university's end-of-semester evaluation and in any mid-semester feedback activity used in this course.

Course Schedule (tentative)

The proposed schedule below is subject to change, and revisions will be posted on Brightspace. Section and page references use the source abbreviations defined under Optional References; the page numbers are printed book pages. The two midterm dates will be filled in by the end of the week.

Week 1: Aug. 24–28

Topics. Statistical learning as risk minimization: statistical models, loss and population risk, empirical risk minimization, inductive bias, realizable and agnostic PAC learning, finite-class bounds, uniform convergence, Hoeffding's inequality, and the ERM transfer argument.

Notes and excerpts. [UML] §§2.1–2.3, pp. 13–20; §§3.1–3.2, pp. 22–27; and §§4.1–4.2, pp. 31–34. [K] §1.1, pp. 1–3.

Week 2: Aug. 31–Sept. 4

Topics. No-Free-Lunch and the bias–complexity tradeoff; approximation versus estimation error; shattering and VC dimension; geometric examples; growth functions; and the Sauer–Shelah–Perles lemma.

Notes and excerpts. [UML] §§5.1–5.2, pp. 37–41; §§6.1–6.3, pp. 43–48; and §6.5.1, pp. 49–50. [GN] §3.6, pp. 212–227.

Week 3: Sept. 7–11

Topics. Ghost samples, effective size, and uniform convergence from growth functions; the qualitative equivalences among finite VC dimension, uniform convergence, ERM learnability, PAC learnability, and agnostic PAC learnability. No class Monday, Sept. 7 (Labor Day).

Notes and excerpts. [UML] §§6.4–6.5, pp. 48–53, especially Theorems 6.7 and 6.11; Chapter 28, §28.1, pp. 341–344.

Week 4: Sept. 14–18

Topics. Quantitative forms of the Fundamental Theorem of Statistical Learning; realizable and agnostic rates; Rademacher symmetrization, Massart's lemma, and contraction; feature maps, the representer theorem, Hilbert-ball complexity, and localized RKHS excess risk.

Notes and excerpts. [UML] Theorem 6.8 and Chapter 28, §§28.1–28.3, pp. 341–350; §§26.1–26.3, pp. 325–335; and §§16.1–16.3, pp. 179–187. [K] §§2.1–2.2, pp. 18–23; §3.7, pp. 54–57; and §5.1, pp. 82–88.

Week 5: Sept. 21–25

Topics. One class meeting is reserved for Midterm 1 (date to be inserted). The new material begins the concentration toolkit: exponential moments, subgaussian and subexponential variables, Orlicz norms, products, Bernstein's inequality, thin-shell norm concentration, and Gaussian Lipschitz concentration.

Notes and excerpts. Midterm 1 scope: [UML] §§2.1–6.5 and Chapter 28, pp. 13–53 and 341–350. New material: [HDP] §2.2 and §§2.6–2.9, pp. 27–48; §3.1, pp. 59–60; and §§5.1–5.2, pp. 140–149. [GN] §§2.2.2–2.2.3, pp. 30–35.

Week 6: Sept. 28–Oct. 2

Topics. Quadratic forms and matrix concentration; Hanson–Wright and matrix Bernstein; volumetric covering and packing; epsilon-net reductions for operator norms; and the operator norm of a subgaussian random matrix.

Notes and excerpts. [HDP] §5.4, pp. 152–159; §§6.1–6.2, pp. 171–178; §§4.1–4.2, pp. 100–112; and §4.4, pp. 115–119. [K] §2.4, pp. 26–31. [Tao] §2.3.1, pp. 107–110.

Week 7: Oct. 5–9

Topics. The Wigner semicircle law by the moment method: trace moments, closed walks, noncrossing pairings, and Catalan numbers; two-sided singular-value bounds, whitening, sample covariance, and PCA. No class Friday, Oct. 9 (Fall Recess).

Notes and excerpts. [Tao] §§2.3.4–2.3.5, pp. 115–124, and §§2.4.1–2.4.2, pp. 137–141. [HDP] §4.6, pp. 122–124, and §4.7, pp. 124–128.

Week 8: Oct. 12–16

Topics. Gaussian comparison and Gaussian width; widths of finite, sparse, ℓ_1, and Euclidean sets; Dudley chaining, entropy integrals, γ_2, and generic chaining; empirical-process and VC bounds through chaining.

Notes and excerpts. [HDP] §§7.1–7.5, pp. 194–212; §8.1, pp. 220–225; and §§8.2–8.5, pp. 226–248. [GN] §2.4.2, pp. 52–59; §3.5.1, pp. 184–194; and §3.6, pp. 212–227.

Week 9: Oct. 19–23

Topics. Matrix deviation on an arbitrary set: subgaussian increments, removal of squares, comparison, and the uniform theorem; applications to effective-rank covariance estimation, infinite-set Johnson–Lindenstrauss embeddings, random sections, and sparse-recovery geometry.

Notes and excerpts. [HDP] §9.1, pp. 260–265; §9.2, pp. 265–269; and §§9.4–9.5, pp. 272–283, especially Theorems 9.1.1–9.1.2, 9.2.2, 9.4.4, and 9.5.1.

Week 10: Oct. 26–30

Topics. One class meeting is reserved for Midterm 2 (date to be inserted). The course then pivots to sampling: the Langevin stochastic differential equation, Itô calculus, the Fokker–Planck equation, stationarity of the target law, synchronous coupling, Wasserstein contraction, and log-Sobolev mixing.

Notes and excerpts. Midterm 2 scope: [HDP] §§2.6–2.9, pp. 39–48; §3.1, pp. 59–60; §§4.2, 4.4, and 4.6–4.7, pp. 105–128. New material: [LV] §§8.1–8.2, pp. 125–133. [GN] §2.5, pp. 60–65.

Week 11: Nov. 2–6

Topics. Geometric Markov-chain Monte Carlo: reversibility, warm starts, conductance, the ball walk, log-concave isoperimetry, and localization; introduction to infinite-dimensional statistical experiments; Gaussian white noise, sequence coordinates, Gaussian shifts, Cameron–Martin geometry, likelihood ratios, and white-noise Kullback–Leibler divergence.

Notes and excerpts. [LV] §§10.1–10.2, pp. 140–153, and §10.5, pp. 154–160. [GN] §§1.1–1.2, pp. 1–12; §§2.6.1–2.6.2, pp. 66–79; and §6.1.1, pp. 467–472.

Week 12: Nov. 9–13

Topics. Multiresolution analysis, compactly supported wavelets, scaling and wavelet coefficients, projection identities, boundary corrections on $[0, 1]$, Besov spaces, coefficient characterizations, embeddings, and projection bias. No class Wednesday, Nov. 11 (Veterans Day).

Notes and excerpts. [GN] §§4.2.1–4.2.3, pp. 305–326, and §4.3.1, pp. 327–337, especially Theorems 4.2.10 and 4.3.2 and Proposition 4.3.8.

Week 13: Nov. 16–20

Topics. Wavelet projection estimators in white noise; stochastic L^p error and bias–variance balance; the minimax upper rate over a Besov ball; testing reductions, Kullback–Leibler control, Fano's inequality, and Varshamov–Gilbert packing.

Notes and excerpts. [GN] §5.1.1, pp. 391–404; §6.3.3, pp. 519–520; Example 3.1.4, p. 115; §6.1.2, pp. 472–475; and §6.3.1, pp. 511–515.

Week 14: Nov. 23–27

Topics. Completion of the matching Besov minimax lower bound: wavelet hypercubes, Besov-ball membership, L^p separation, KL control, and Fano testing. No class Wednesday or Friday (Thanksgiving recess).

Notes and excerpts. [GN] §4.3.1, pp. 327–337, and §§6.3.1–6.3.3, pp. 511–520, especially Theorems 6.3.2 and 6.3.8.

Week 15: Nov. 30–Dec. 4

Topics. Adaptation by Lepski's method; oracle resolutions, stochastic thresholds, adaptive L^2 and L^∞ estimation, and thresholding; the obstruction to honest adaptation on nested Hölder balls, separation and self-similarity; course synthesis and review.

Notes and excerpts. [GN] §§8.2.1–8.2.2, pp. 614–627; §8.3.1, pp. 628–636; and §8.3.3, pp. 644–657, especially Theorems 8.2.2, 8.2.5, 8.2.7–8.2.8, 8.3.1, and 8.3.15–8.3.16.

Final assessment period: Dec. 16, 11:00 AM–1:00 PM

Final examination or finalproject submission/presentation, according to the assessment format selected by the student.

Statement on University Academic and Support Systems

Students and Disability Accommodations
USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) determines appropriate accommodations for students who encounter disability-related barriers. After a student completes the OSAS process and accommodations are approved, the student must provide the Letter of Accommodation to the instructor and follow up with a discussion as early as possible. Accommodations are not retroactive. Contact OSAS at (213) 740-0776 or osasfrontdesk@usc.edu.

Student Financial Aid and Satisfactory Academic Progress
Eligibility for certain forms of financial aid requires satisfactory academic progress toward degree objectives. The Financial Aid Office provides undergraduate and graduate requirements and appeal procedures.

Support Systems
Counseling and Mental Health – (213) 740-9355 (WELL), available 24/7. Free and confidential mental health care includes short-term psychotherapy, group counseling, workshops, and crisis intervention.

988 Suicide and Crisis Lifeline – call or text 988, available 24/7. The Lifeline provides free and confidential emotional support to people in suicidal crisis or emotional distress.

CARE-SC: Confidential Advocacy, Resources, and Education Support Center – (213) 740-9355 (WELL), available 24/7/365. Confidential advocates and counselors support students affected by gender- and power-based harm.

Office of Civil Rights Compliance – (213) 740-5086. Information about reporting and support related to harassment, discrimination, and retaliation.

USC Campus Support and Intervention – (213) 740-0411. Assistance for students and families facing complex personal, financial, or academic issues.

USC Emergency Information. Current safety updates and continuity-of-instruction information.

USC Department of Public Safety. Emergency: UPC (213) 740-4321; HSC (323) 442-1000. Non-emergency: UPC (213) 740-6000; HSC (323) 442-1200.

Office of the Ombuds – (213) 821-9556 (UPC) or (323) 442-0382 (HSC). A confidential place to discuss USC-related concerns and explore options.

Occupational Therapy Faculty Practice – (323) 442-2850. Lifestyle Redesign services supporting health-promoting habits, quality of life, and academic performance.

Office of Sustainability. Resources and opportunities for studying, working, and living sustainably at USC.

USC Report & Response. A central portal for reporting concerns, including anonymous reports, and connecting with university support. Questions may be directed to the Office of Professionalism and Ethics at ope@usc.edu.